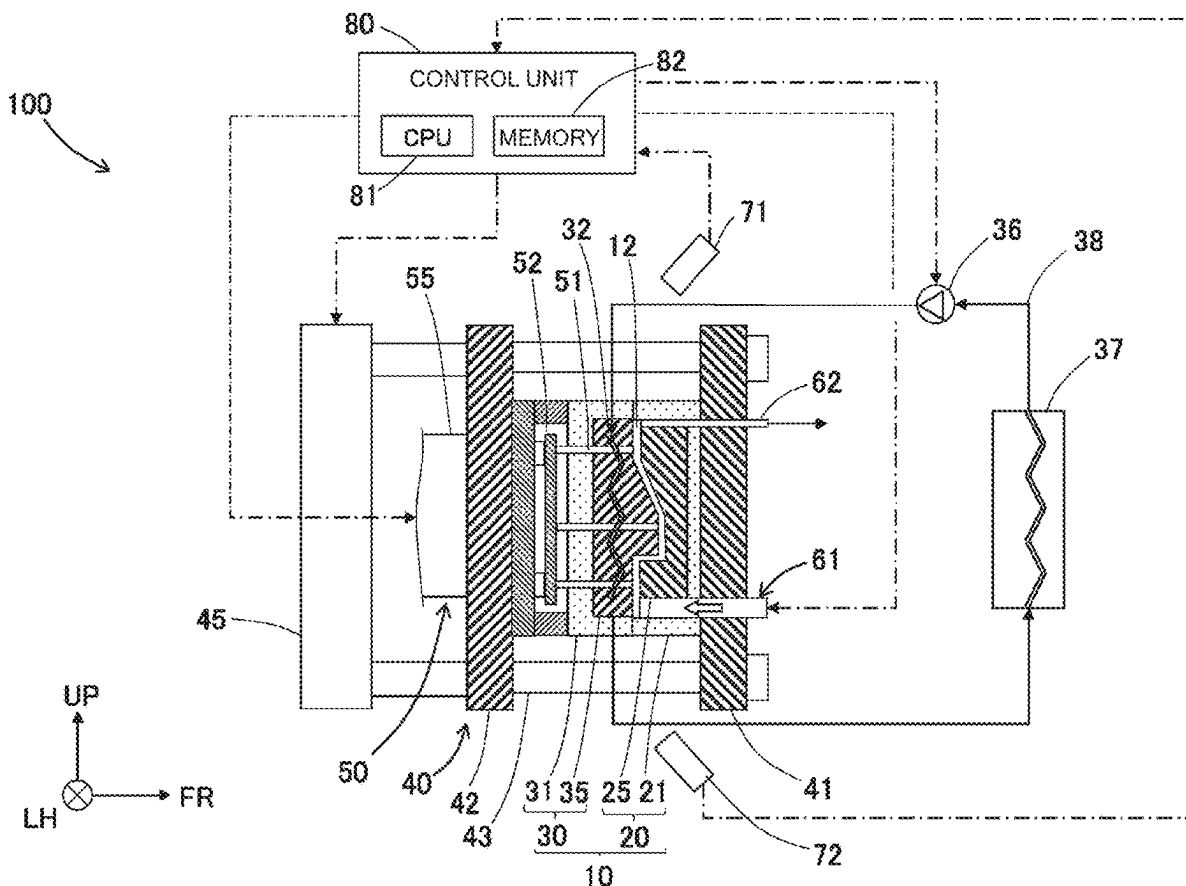


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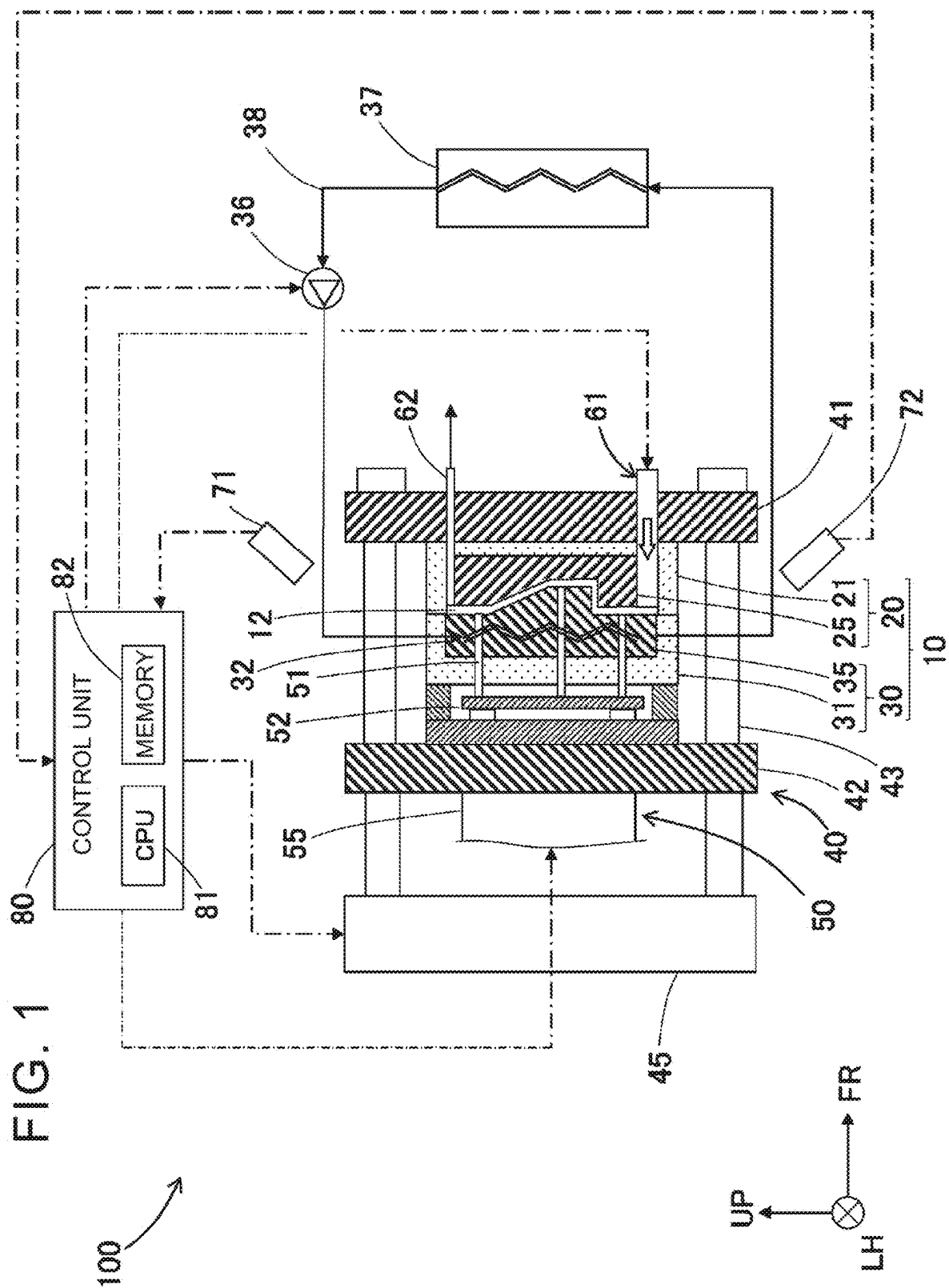
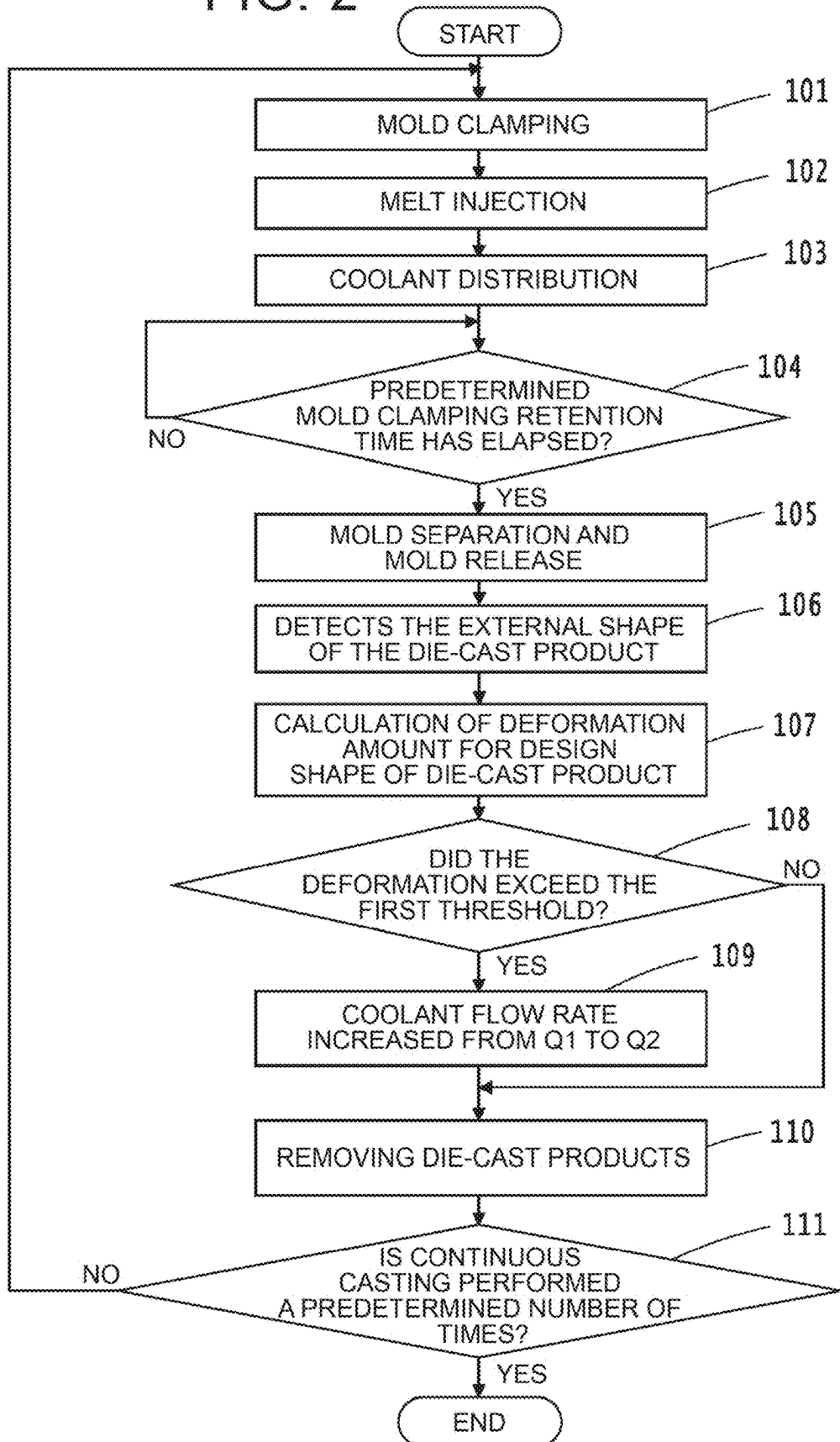


FIG. 2



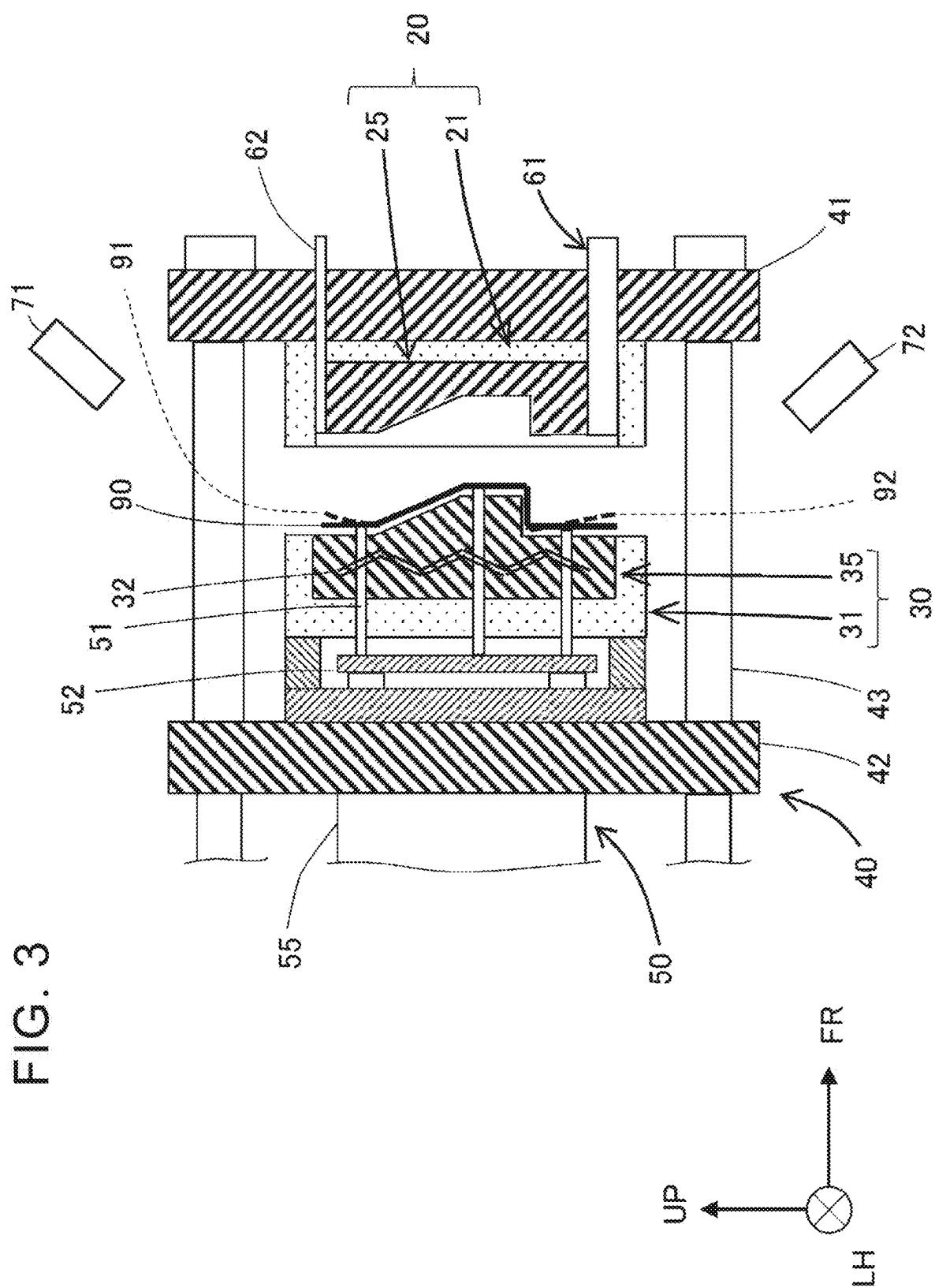


FIG. 4

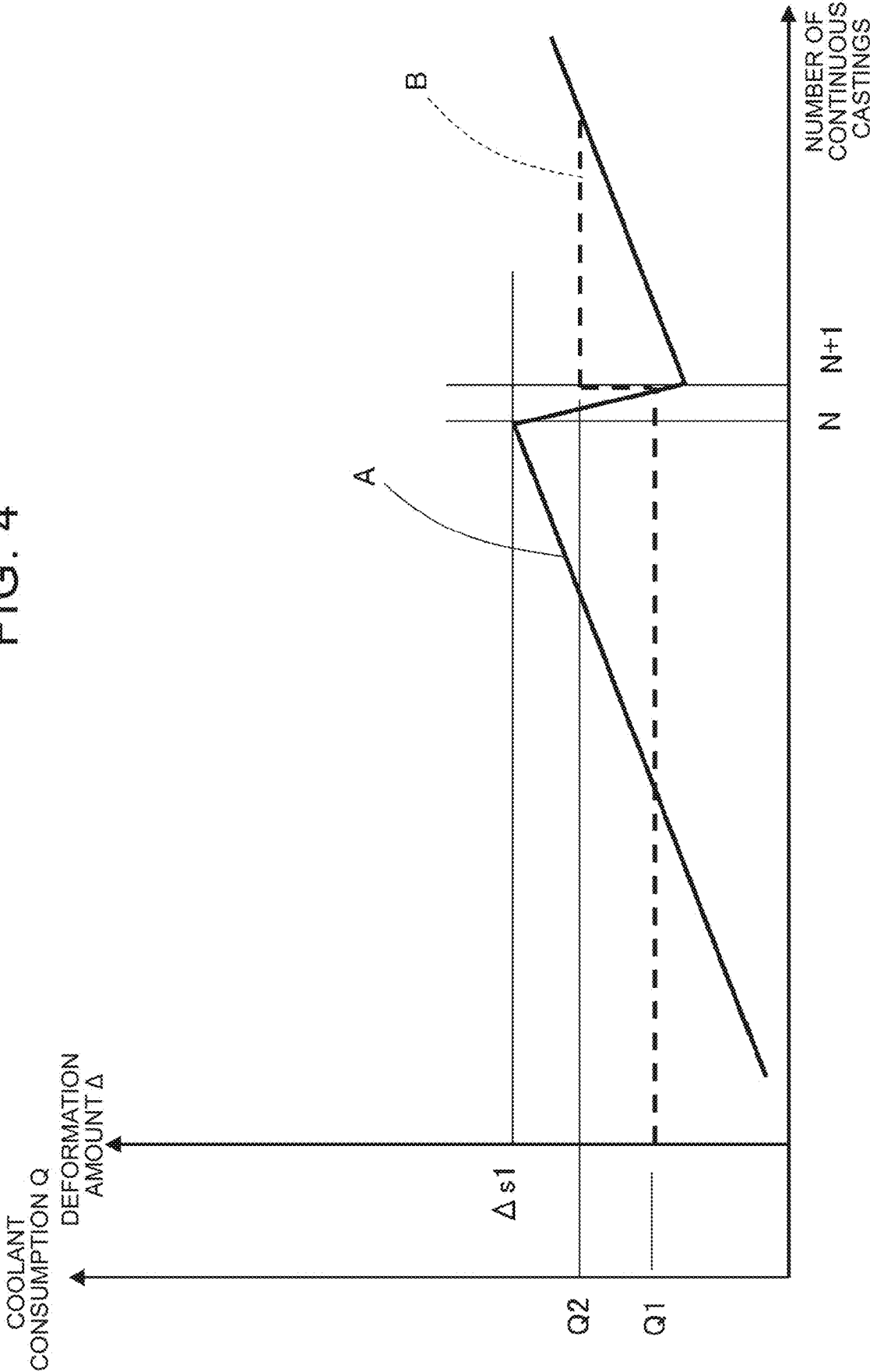


FIG. 5

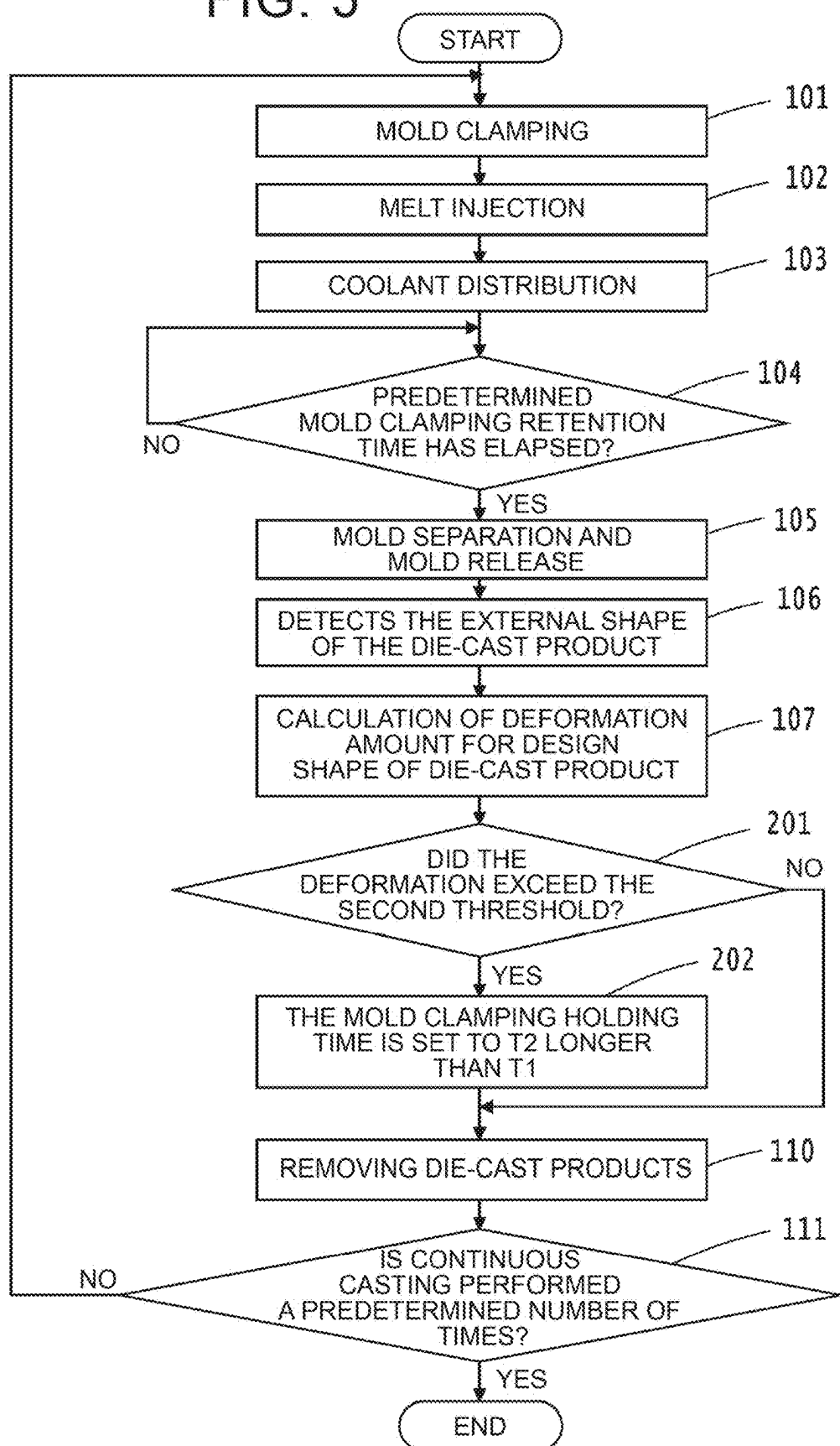
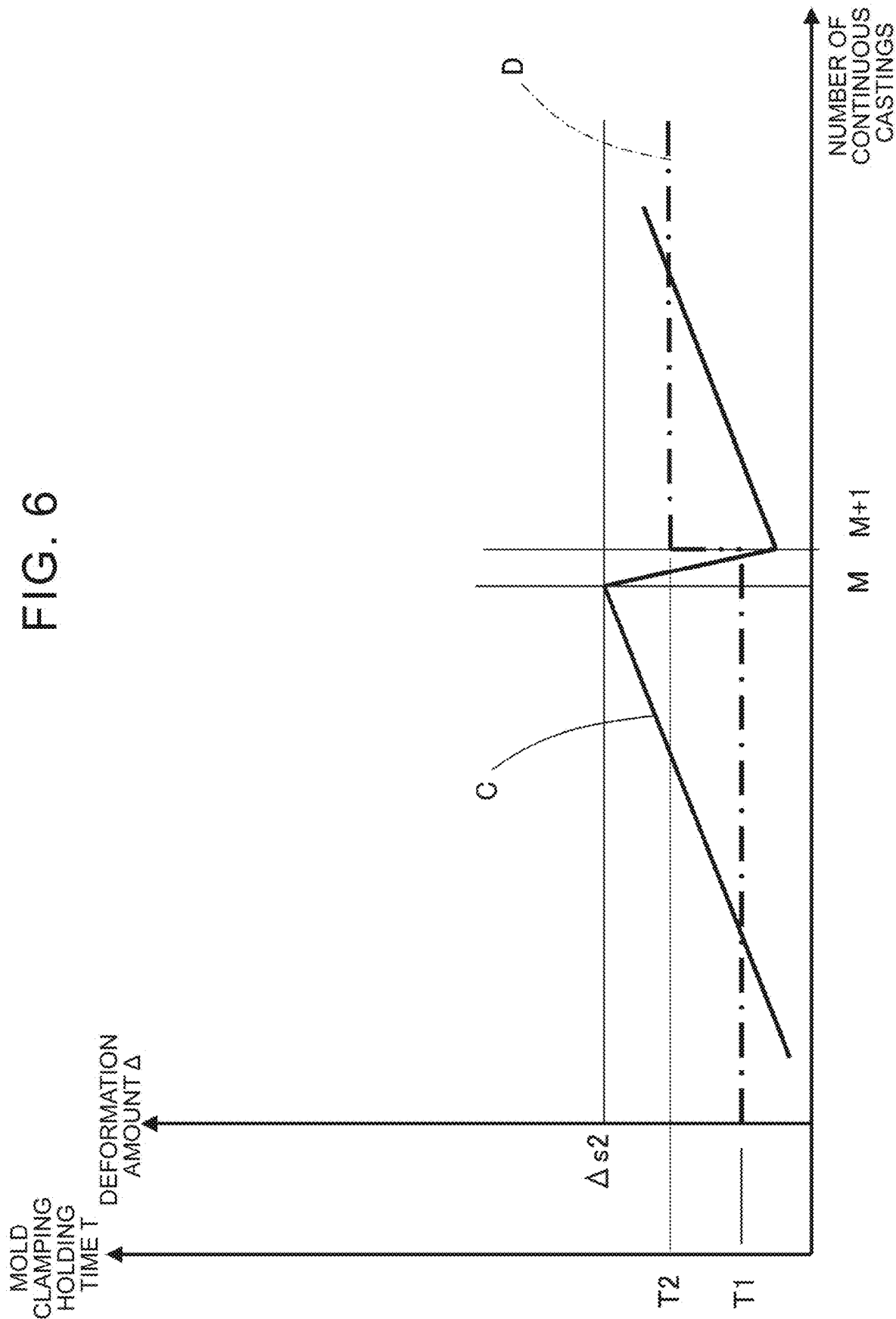


FIG. 6



CASTING APPARATUS AND CASTING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031546 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a structure of a casting apparatus and a casting method using the casting apparatus.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2003-191065 (JP 2003-191065 A) discloses a casting apparatus capable of efficiently replacing a dedicated part by drawing out the dedicated part from a general-purpose part and attaching a new dedicated part.

SUMMARY

[0004] Now, when die-cast products are consecutively cast using the casting apparatus such as described in JP 2003-191065 A, deformation of the die-cast product may become great. However, in the casting apparatus according to the related art, feedback of deformation amount to casting conditions cannot be performed, since the deformation amount with respect to a designed shape of the die-cast product is not detected. Accordingly, in the casting apparatus according to the related art, suppressing increase in the deformation amount when performing consecutive casting is difficult.

[0005] Accordingly, an object of the present disclosure is to suppress increase in the deformation amount of a die-cast product when performing consecutive casting of the die-cast product.

[0006] The casting apparatus according to the present disclosure is

[0007] a casting apparatus that performs casting of a die-cast product, the casting apparatus including

[0008] a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel,

[0009] a mold clamping device for performing mold clamping and separation of the mold,

[0010] a coolant pump for causing coolant to flow over the coolant channel of the mold,

[0011] a shape detector for detecting an external shape of the die-cast product, and

[0012] a control unit for adjusting mold clamping holding time of the mold by the mold clamping device or a coolant flow rate of the coolant pump, in which

[0013] the control unit includes a processor for performing information processing, and the processor

[0014] acquires the external shape of the die-cast product from the shape detector when separating the mold following casting,

[0015] calculates a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is acquired, and

[0016] adjusts the mold clamping holding time or the coolant flow rate based on the deformation amount that is calculated.

[0017] In this way, the mold clamping holding time or the coolant flow rate is adjusted in accordance with the deformation amount with respect to the design shape of the die-cast product following casting, and accordingly increase in the deformation amount of the die-cast product can be suppressed when performing consecutive casting.

[0018] In the casting apparatus according to the present disclosure, the processor may increase the coolant flow rate when the deformation amount that is calculated exceeds a predetermined threshold value.

[0019] Accordingly, rise in the temperature of the die-cast product when performing consecutive casting can be suppressed, and increase in the deformation amount of the die-cast product can be suppressed.

[0020] In the casting apparatus according to the present disclosure, the processor may increase the mold clamping holding time when the deformation amount that is calculated exceeds a predetermined other threshold value.

[0021] During the mold clamping holding time, the coolant is distributed to the mold, and accordingly rise in the temperature of the die-cast product when performing consecutive casting due to increasing the mold clamping holding time can be suppressed, and increase in the deformation amount of the die-cast product can be suppressed.

[0022] In the casting apparatus according to the present disclosure, the shape detector may be a laser scanner or an ultrasonic shape detector.

[0023] Thus, the deformation amount of the die-cast product can be detected non-contactlessly.

[0024] A method of casting a die-cast product according to the present disclosure includes

[0025] preparing a casting apparatus that includes a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel, a mold clamping device for performing mold clamping and separation of the mold, a coolant pump for causing coolant to flow over the coolant channel of the mold, and a shape detector for detecting an external shape of the die-cast product,

[0026] detecting the external shape of the die-cast product by the shape detector when separating the mold following casting,

[0027] calculating a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is detected, and

[0028] increasing a coolant flow rate of the coolant pump when the deformation amount that is calculated exceeds a predetermined threshold value, or increasing the mold clamping holding time of the mold when the deformation amount that is calculated exceeds a predetermined other threshold value.

[0029] Thus, increase in the deformation amount of the die-cast product when performing consecutive casting can be suppressed.

[0030] The present disclosure can suppress increase in the deformation amount of the die-cast product when the die-cast product is consecutively cast.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0032] FIG. 1 is a system diagram illustrating a configuration of a casting apparatus according to an embodiment;

[0033] FIG. 2 is a flow chart illustrating the operation of the casting apparatus shown in FIG. 1 in casting die-cast products;

[0034] FIG. 3 is an explanatory view showing a casting apparatus in a state in which a mold is separated after casting;

[0035] FIG. 4 is a diagram illustrating a change in the deformation amount of the die-cast product and a change in the coolant flow rate with respect to the number of times of continuous casting;

[0036] FIG. 5 is a flow chart showing another operation of the casting apparatus shown in FIG. 1 in casting die-cast products; and

[0037] FIG. 6 is a diagram illustrating a change in the deformation amount of the die-cast product and a change in the mold clamping holding time with respect to the number of times of continuous casting.

DETAILED DESCRIPTION OF EMBODIMENTS

[0038] Hereinafter, the casting apparatus 100 of the embodiment will be described with reference to the drawings. As illustrated in FIG. 1, the casting apparatus 100 includes a mold 10, a mold clamping device 40, an extrusion device 50, an injection device 61, an exhaust pipe 62, a coolant pump 36, laser scanners 71 and 72, and a control unit 80. In the drawings, reference numerals FR, LH, UP denote the front, left, and upper sides of the casting apparatus 100, respectively. The opposing orientations of FR, LH, UP are rearward, rightward, and downward, respectively.

[0039] The mold clamping device 40 includes a fixed plate 41, a movable platen 42, a tie bar 43, and a driving unit 45. The fixed plate 41 is fixed to a distal end portion of a tie bar 43 which is a rod member. The movable platen 42 is disposed so as to face the fixed plate 41, and moves forward along the tie bar 43 by the driving unit 45.

[0040] A mold 10 is attached to the mold clamping device 40. The mold 10 includes a fixed mold 20 and a movable mold 30. A fixed mold 20 is attached to the fixed plate 41, and a movable mold 30 is attached to the movable platen 42. The mold clamping device 40 opens and closes the mold 10 by moving the movable mold 30 in the front-rear direction together with the movable platen 42. Then, the movable mold 30 is moved forward and pressed against the fixed mold 20, whereby the mold clamping of the mold 10 is performed. FIG. 1 shows a mold 10 in a clamped state by a mold clamping device 40. When mold clamping of the mold 10 is performed, a cavity 12 is formed between the fixed mold 20 and the movable mold 30. The cavity 12 is a portion of the internal space of the mold 10 having a shape corresponding to the shape of the die-cast product 90 (see FIG. 3). Here, the die-cast product 90 may be, for example, a vehicle body structural member.

[0041] The injection device 61 pumps the molten metal, which is the material of the die-cast product 90, into the cavity 12 of the mold 10 as indicated by the white arrow in

FIG. 1. Further, the air in the cavity 12 is exhausted from the exhaust pipe 62 by a vacuum device (not shown) (see an arrow in FIG. 1). The extrusion device 50 includes an extrusion pin 51, an extrusion plate 52, and an extrusion driving unit 55. The extrusion device 50 separates the die-cast product 90 from the movable mold 30 by projecting the extrusion pins 51 when the mold 10 is separated.

[0042] The fixed mold 20 includes a main mold 21 and an insert 25. The movable mold 30 includes a main mold 31 and an insert 35. Here, the inserts 25 and 35 are portions of the fixed mold 20 and the movable mold 30 that are replaced according to the die-cast product 90 to be molded. The main molds 21 and 31 are portions fixed to the fixed plate 41 and the movable platen 42 of the mold clamping device 40, and are portions commonly used for the die-cast product 90 to be molded.

[0043] A coolant channel 32 is provided inside the insert 35 of the movable mold 30. In the coolant channel 32, the coolant pump 36 and the coolant cooler 37 are connected by a coolant pipe 38. The coolant pressurized by the coolant pump 36 flows into the coolant channel 32 of the insert 35. The coolant whose temperature has risen in the coolant channel 32 is cooled by the coolant cooler 37 and circulated to the coolant pump 36. In this way, the coolant pump 36 allows the coolant to flow through the coolant channel 32 of the insert 35.

[0044] Laser scanners 71 and 72 are disposed slightly behind the fixing plate 41. The laser scanners 71 and 72 are shape detectors that irradiate the die-cast product 90 (see FIG. 3) with laser light and detect the outer shape of the die-cast product 90 by reflected waves. The laser scanner 71 detects an outer shape of an upper portion of the die-cast product 90, and the laser scanner 72 detects an outer shape of a lower portion of the die-cast product 90.

[0045] The control unit 80 is a computer including a CPU 81 that is a processor that performs information processing, and a memory 82 that stores a control program and control data. The driving unit 45 of the mold clamping device 40, the extrusion driving unit 55 of the extrusion device 50, the injection device 61, and the coolant pump 36 are connected to the control unit 80 and operate according to an instruction from the control unit 80. Further, the laser scanners 71 and 72 are connected to the control unit 80. The detected external shape data of the die-cast product 90 is input to the control unit 80. As described with reference to FIGS. 2 to 6, the control unit 80 adjusts the mold clamping holding time T by the mold clamping device 40 or the coolant flow rate Q of the coolant pump 36.

[0046] Next, with reference to FIGS. 2 to 4, the operation of the casting apparatus 100 when casting the die-cast product 90 will be described. As shown in FIG. 1, a mold 10 is set in the casting apparatus 100.

[0047] As shown in step 101 of FIG. 2, CPU 81 of the control unit 80 moves the movable mold 30 forward together with the movable platen 42 by operating the driving unit 45 of the mold clamping device 40, the mold clamping of the mold 10 is pressed against the fixed mold 20. Then, in step 102 of FIG. 2, CPU 81 of the control unit 80 operates the injection device 61 to inject the molten metal into the cavity 12 of the mold 10. Then, in step 103 of FIG. 2, CPU 81 operates the coolant pump 36 to cause the coolant to flow through the coolant channel 32. Then, CPU 81 maintains the duty of the coolant pump 36 at a predetermined value, maintains the coolant flow rate Q at a predetermined flow

rate, and maintains the mold clamping state. CPU 81 waits until the predetermined mold clamping holding time T elapses, as shown in step 104 of FIG. 2. Then, when the predetermined mold clamping holding time T has elapsed, CPU 81 determines YES in step 104 of FIG. 2 and proceeds to step 105 of FIG. 2.

[0048] In step 105 of FIG. 2, CPU 81 operates the driving unit 45 of the mold clamping device 40 as shown in FIG. 3 to move the movable mold 30 backward together with the movable platen 42, and separates the mold 10 into the fixed mold 20 and the movable mold 30. When the mold 10 is separated after casting, a die-cast product 90 is formed in the region of the cavity 12 shown in FIG. 1 (see FIG. 3). Further, when the mold 10 is separated into the fixed mold 20 and the movable mold 30, the laser beam from the laser scanners 71 and 72 can reach the surface of the die-cast product 90. In step 105 of FIG. 2, CPU 81 operates the extrusion driving unit 55 of the push-out device 50 to move the push-out pin 51 forward to release the die-cast product 90 from the movable mold 30.

[0049] In step 106 of FIG. 2, CPU 81 detects the outer shape of the die-cast product 90 by the laser scanners 71 and 72, and acquires data of the detected outer shape from the laser scanners 71 and 72. At this time, CPU 81 acquires the external shapes of the outer peripheral portions 91 and 92 of the die-cast product 90 in which the deformation Δ from the designed shape increases. Then, in step 107 of FIG. 2, CPU 81 compares the outer shape of the die-cast product 90 acquired from the laser scanners 71 and 72 with the design shape, and calculates the deformation Δ with respect to the design shape as the difference.

[0050] In step 108 of FIG. 2, CPU 81 determines whether the deformation Δ exceeds the first threshold $\Delta s1$. Then, if it is determined YES in step 108 of FIG. 2, the process proceeds to step 109 of FIG. 2, and the coolant flow rate Q at the time of the next casting is increased to a Q2 larger than Q1 of the initialization. On the other hand, if CPU 81 determines NO in step 108 of FIG. 2, it skips step 109 of FIG. 2 and proceeds to step 110 of FIG. 2. Then, in step 110 of FIG. 2, CPU 81 operates a robot (not shown) to remove the die-cast product 90 from the movable mold 30.

[0051] In step 111 of FIG. 2, CPU 81 determines whether or not casting has been performed continuously a predetermined number of times. If CPU 81 determines YES in step 111 of FIG. 2, the casting process is stopped. On the other hand, if it is determined NO in step 111 of FIG. 2, CPU 81 returns to step 101 of FIG. 2 to begin the next casting.

[0052] Next, with reference to FIG. 4, a change in the deformation amount Δ and the coolant flow rate Q of the die-cast product 90 with respect to the number of times of continuous casting will be described. In FIG. 4, a solid line A indicates a change in the deformation amount Δ , and a broken line B indicates a change in the coolant flow rate Q.

[0053] As shown in FIG. 4, when the first casting is performed, the coolant flow rate Q is a default Q1. When the first casting is completed, the temperature of the die-cast product 90 at the end of the casting is not very high, and the deformation amount Δ is also small.

[0054] As the number of times of continuous casting increases, the temperature of the die-cast product 90 at the end of casting gradually increases, and as shown by the solid line A in FIG. 4, the deformation amount Δ gradually increases. Then, during the N-th casting, the deformation amount Δ exceeds the first threshold value $\Delta s1$. At this time,

as indicated by a broken line in FIG. 3, the outer peripheral portions 91 and 92 of the die-cast product 90 are deformed. Then, CPU 81 determines YES in step 108 of FIG. 2, and increases the coolant flow rate Q to a Q2 larger than the default Q1 in step 109 of FIG. 2.

[0055] At the time of the N+1-th casting, CPU 81 makes the duty of the coolant pump 36 larger than the initial setting in step 103 of FIG. 2, and makes the coolant flow rate Q of the coolant channel 32 Q2 (see the broken line B in FIG. 4). As a result, in the N+1-th casting, the temperature of the die-cast product 90 is lower than that in the N-th casting when the mold clamping holding is performed in step 104 of FIG. 2. Therefore, as shown by the solid line A in FIG. 4, in the N+1-th casting, the deformation amount Δ is smaller than the first threshold value $\Delta s1$.

[0056] Thereafter, as the number of times of continuous casting increases, the deformation amount Δ increases again. CPU 81 again increases the coolant flow rate Q when the deformation Δ exceeds the first threshold $\Delta s1$.

[0057] As described above, in the casting apparatus 100, when the deformation Δ exceeds a predetermined threshold, CPU 81 increases the coolant flow rate Q. Therefore, an increase in the temperature of the die-cast product 90 during continuous casting can be suppressed, and an increase in the deformation amount Δ of the die-cast product 90 can be suppressed.

[0058] Next, with reference to FIG. 5 and FIG. 6, another operation of the casting

[0059] apparatus 100 when casting the die-cast product 90 will be described. Another operation is an operation of increasing the mold clamping holding time T instead of increasing the coolant flow rate Q when the deformation amount Δ exceeds the second threshold value $\Delta s2$. The same operations as those described above with reference to FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. The second threshold value $\Delta s2$ is another threshold value different from the first threshold value $\Delta s1$.

[0060] As shown in FIG. 5, in another operation, CPU 81 determines whether the deformation Δ exceeds the second threshold $\Delta s2$ in step 201 of FIG. 5. If CPU 81 determines YES in step 201 of FIG. 5, the process proceeds to step 202 of FIG. 5, and the mold clamping holding time T at the time of the next casting is set to T2 longer than the default T1.

[0061] With reference to FIG. 6, the variation of the deformation amount Δ and the mold clamping holding time T of the die-cast product 90 with respect to the number of times of continuous casting in another operation will be described. In FIG. 6, a solid line C indicates a change in the deformation amount Δ , and a dashed-dotted line D indicates a change in the mold clamping holding time T.

[0062] As shown in FIG. 6, when the deformation amount Δ exceeds the second threshold $\Delta s2$ at the number of times of casting of the M-th time, CPU 81 sets the mold clamping holding time T to T2 longer than the default T1 in step 202 of FIG. 5 (see the dashed-dotted line D in FIG. 6). Thus, in the M+1-th casting, CPU 81 holds the mold clamping for T2 longer than the default T1 by the mold clamping device 40, so that the temperature of the die-cast product 90 is lower than that during the M-th casting. Therefore, as shown by the solid line C in FIG. 6, in the M+1-th casting, the deformation amount Δ is smaller than the second threshold value $\Delta s2$. As

a result, it is possible to suppress an increase in the deformation amount Δ of the die-cast product **90** during continuous casting.

[0063] In the above explanation, in other operations, CPU **81** has been explained as increasing the mold clamping holding time T instead of increasing the coolant flow rate Q . However, without being limited thereto, CPU **81** may be performed by combining an increased coolant flow rate Q and an extended mold clamping holding time T .

[0064] Further, in the above description, the outer shape of the die-cast product **90** is detected by the laser scanners **71** and **72**. However, the present disclosure is not limited thereto, and for example, an ultrasonic shape detector that detects the outer shape of the die-cast product **90** by ultrasonic waves may be used. As a result, the outer shape of the die-cast product **90** can be detected in a non-contact manner.

[0065] In the above description, the coolant channel **32** is provided in the insert **35** of the movable mold **30**, but the present disclosure is not limited thereto. For example, a coolant channel may be provided in the insert **25** of the fixed mold **20** to allow the coolant to flow through the coolant pump **36**. In addition, a coolant channel may be provided in both the insert **35** of the movable mold **30** and the insert **25** of the fixed mold **20** to allow the coolant to flow there-through.

[0066] Each step of the casting apparatus **100** described with reference to FIG. **2** is also a casting method performed by the casting apparatus **100**. In this case, steps **105** and **106** of FIG. **2** are steps of detecting the outer shape of the die-cast product **90** by the laser scanners **71** and **72** when the mold **10** is separated after casting. Step **107** of FIG. **2** is a step of calculating the deformation amount Δ with respect to the design shape of the die-cast product **90** based on the outer shape of the die-cast product **90**. Further, steps **108** to **109** of FIG. **2** are steps of increasing the coolant flow rate Q of the coolant pump **36** when the deformation amount Δ exceeds a predetermined threshold value. Steps **201** to **202** in FIG. **5** are steps of increasing the mold clamping holding time T when the deformation amount Δ exceeds a predetermined threshold value.

What is claimed is:

1. A casting apparatus that performs casting of a die-cast product, the casting apparatus comprising:

- a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel;
- a mold clamping device for performing mold clamping and separation of the mold;
- a coolant pump for causing coolant to flow over the coolant channel of the mold;
- a shape detector for detecting an external shape of the die-cast product; and

a control unit for adjusting mold clamping holding time of the mold by the mold clamping device or a coolant flow rate of the coolant pump, wherein

the control unit includes a processor for performing information processing, and
the processor

acquires the external shape of the die-cast product from the shape detector when separating the mold following casting,

calculates a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is acquired, and

adjusts the mold clamping holding time or the coolant flow rate based on the deformation amount that is calculated.

2. The casting apparatus according to claim 1, wherein the processor increases the coolant flow rate when the deformation amount that is calculated exceeds a predetermined threshold value.

3. The casting apparatus according to claim 2, wherein the processor increases the mold clamping holding time when the deformation amount that is calculated exceeds a predetermined other threshold value.

4. A casting apparatus according to claim 1, wherein the shape detector is a laser scanner or an ultrasonic shape detector.

5. A casting method of a die-cast product, the casting method comprising:

preparing a casting apparatus that includes a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel, a mold clamping device for performing mold clamping and separation of the mold, a coolant pump for causing coolant to flow over the coolant channel of the mold, and a shape detector for detecting an external shape of the die-cast product;

detecting the external shape of the die-cast product by the shape detector when separating the mold following casting;

calculating a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is detected; and

increasing a coolant flow rate of the coolant pump when the deformation amount that is calculated exceeds a predetermined threshold value, or increasing the mold clamping holding time of the mold when the deformation amount that is calculated exceeds a predetermined other threshold value.

* * * * *